

Hasun Lee

M.S. IN MECHANICAL ENGINEERING

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Education

Seoul National University

M.S. IN MECHANICAL ENGINEERING

Seoul, Korea

Mar. 2019 - Aug. 2021

- Interactive & Networked Robotics Laboratory (INRoL) (Advisor: Prof. Dongjun Lee)
- Thesis: Contact Wrench Estimation for Large-Size Operational Aerial Skeleton System with Distributed IMUs

Pohang University of Science and Technology (POSTECH)

B.S. IN PHYSICS

Pohang, Korea

Mar. 2014 - Feb. 2019

- Graduated with *Magna Cum Laude*
- Total GPA: 3.73/4.3
- Thesis: 3D printed Watt balance device and mass measuring with electro-magnetic force

Seoul Science High School

GRADUATED

Seoul, Korea

Mar. 2011 - Feb. 2014

- Early Entrance
- Student Body Vice President

Work Experience

StradVision Corp.

COMPUTER VISION ENGINEER

Seoul, S.Korea

Sep. 2021 - Present

Joined prior to the company's KOSDAQ IPO (expected Jun. 2026); contributing through pre-IPO growth phase.

- Worked on an LSS-based 3D detector for multi-fisheye cameras, under active vehicle testing for a confirmed production project.
 - Set up and modified an LSS/BEVDet training framework for multi-fisheye inputs.
- Built a 3D Asset Harvesting pipeline combining 2D / 3D generation and segmentation models.
 - Lifted production-homologation Recall for parking stoppers and pedestrians from 0.20 to 0.997.
- Applied data augmentation to real RGB images for improved model generalization.
 - Augmented parking-slot and nearby-vehicle samples using Depth-Pro and SAM2.
- Developed a pipeline for fisheye image synthesis and data acquisition in the CARLA simulator.
 - Generated a synthetic dataset of fisheye RGB, depth, and 3D-label frames.

Publications

Y.Kim, **H.Lee**, J.Lee, and D. J. Lee, Wrench Estimation of Modular Manipulator with External Actuation and Joint Locking, INTERNATIONAL CONFERENCE ON INTELLIGENT ROBOTS AND SYSTEMS (IROS).

Oct. 2023

H.Lee, Y.Kim, H.Yang, and D. J. Lee, Development of Downsized LASDRA with 2-DoF Joint Locking Device, AERIAL ROBOTIC SYSTEMS PHYSICALLY INTERACTING WITH THE ENVIRONMENT (AIRPHARO).

Oct. 2021

H.Lee, Y.Kim, H.Yang, and D. J. Lee, Preliminary Results on Downsized LASDRA System with Joint Locking, UBIQUITOUS ROBOTS (UR).

Jul. 2021

S.Park, **H.Lee** and D. J. Lee, Robust Motion Control of Robotic Systems with Environmental Interaction via Data-Driven Inversion of CPG, INTERNATIONAL CONFERENCE ON CONTROL, AUTOMATION AND SYSTEMS (ICCAS).

Oct. 2020

S.Park, **H.Lee** and D. J. Lee, Preliminary Results on Motion Control of Robotic Systems with Environmental Interaction via Data-Driven Inversion of CPG, KOREA ROBOTICS SOCIETY ANNUAL CONFERENCE (KROC).

Aug. 2020

H.Lee, S.Park and D. J. Lee, Variable Speed Control Moment Gyro's Gimbal Arrangement Optimization and Control Allocation Improving, ICROS ANNUAL CONFERENCE.

Jul. 2020

Projects

Auto Parking Software Mass Production Development

CORE ENGINEER

StradVision

Sep. 2021 - Present

- Developed algorithms for parking space recognition and obstacle detection for automatic parking.
- Served as the main contributor for the object detection feature.

Very Long, Slender and Light Robot Arm with Distributed Rotor Actuation for Aerial Operation and Manipulation

LEADING STUDENT RESEARCHER

National Research Foundation of Korea

Apr. 2020 - Mar. 2021

- Development of joint locking LASDRA system.
- External wrench estimation with multiple IMU sensor fusion using UKF.

Development of Attitude Control Algorithm for Satellite considering Singularity Avoidance of CMG Cluster

LEADING STUDENT RESEARCHER

Justek Inc.

Apr. 2019 - Mar. 2020

- Newton-Euler dynamic analysis and control of variable speed control moment gyro (CMG).
- Design optimization of CMG cluster by data-driven method.

Teaching Experience

Spring
2020

Teaching Assistant, Undergraduate Course: Mechanical System Modeling and Control

Seoul National University

Honors & Awards

2010, 2011 **1st Prize**, International Astronomy Olympiad (IAO)

Ukraine,
Kazakhstan

2014 - 2018 **Scholarship**, National Science & Technology Scholarship (NSTS)

Korea Student Aid
Foundation

Extracurricular Activity

INTERNSHIP

July 2017 **Financial Engineering team**, Price determining of ELS products by Monte-Carlo method

NH Investment &
Securities Co., Ltd.

EXCHANGE STUDENT

2016 **National University of Singapore**, Exchange student for 1 Semester

Singapore

Skills

Programming	Python, C/C++, Matlab, ROS, LabVIEW
Library	PyTorch, OpenCV, NumPy, ONNX
DevOps	Git, GitHub, Docker
Simulator	Carla, Gazebo
Design	SolidWorks, Onshape
OS	Linux, Windows
Languages	Korean, English

Courses

Undergraduate	System Control, Sensors and Measurements, Programming and Problem Solving, Computational Physics, Mathematical Physics
Graduate	Linear and Nonlinear System Control, Robot Mechanics, Reinforcement Learning, Advanced Filtering, Convex Optimization